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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester B.Tech (CBCS) Degree Examinations, December 2025

25MAT102 – MATRIX ALGEBRA AND CALCULUS

(For ACT, ECE, EEE, VDT)

Duration: 3 Hours]

[Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each Unit.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

CO1: Q1 to Q4; CO2: Q5 to Q8; CO3: Q9 to Q12; CO4: Q13 to Q16; CO5: Q17 to Q20.

- If every minor of order 3 of matrix A is zero, then rank of A is
 - 3
 - Less than 3
 - 0
 - None of these
- If A is 3×4 matrix the maximum possible value of the rank A is
 - 4
 - 3
 - 2
 - 0
- If the rank of a matrix A is 3 then the rank of transpose of A is
 - Less than 3
 - Equal to 3
 - Greater than 3
 - Equal to 0
- If rank of a coefficient matrix and rank of the augmented matrix are equal then system is
 - Inconsistent
 - Consistent
 - Equal
 - None
- The power method estimates
 - All eigenvalues
 - All eigenvectors
 - Determinant
 - Dominant eigenvalue
- If 5 is one of the eigenvalues of A , then 25 is an eigenvalue of:
 - $A + I$
 - A^2
 - A^{-1}
 - $2A$
- In Jacobi's method, the rotation angle θ typically determined:
 - Based on the determinant
 - From the trace of the matrix
 - Arbitrarily chosen small value
 - Using the formula $\tan(2\theta) = \frac{2a_{ij}}{a_{ii} - a_{jj}}$
- For $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 3 & 2 & 1 \end{bmatrix}$, when a Givens' rotation is applied to eliminate $A_{1,3} = 3$ while maintaining symmetry, another element that becomes zero automatically is _____.
 - $A_{3,1}$
 - $A_{1,2}$
 - $A_{2,3}$
 - $A_{2,2}$

9. The derivative of arc S for the curve $x = f(t), y = g(t)$ is $\frac{ds}{dt} =$ _____
- A) $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$ B) $\sqrt{\left(\frac{dx}{dt}\right)^2 - \left(\frac{dy}{dt}\right)^2}$
- C) $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$ D) $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$
10. The radius of curvature for the curve $r = f(\theta)$ is $\rho =$ _____
- A) $\left(\frac{(r^2 + (r_1)^2)^{3/2}}{r^2 + 2(r_1)^2 - rr_2}\right)$ B) $\left(\frac{(r^2 + (r_1)^2)^{2/3}}{r^2 + 2(r_1)^2 - rr_2}\right)$
- C) $\left(\frac{(r^2 + (r_1)^2)^{3/2}}{r^2 - 2(r_1)^2 - rr_2}\right)$ D) $\left(\frac{(r^2 + (r_1)^2)^{3/2}}{r^2 + 2(r_1)^2 + rr_2}\right)$
11. Two polar curves intersect orthogonally if _____
- A) $\tan\phi_1 \cdot \tan\phi_2 = \frac{\pi}{2}$ B) $\tan\phi_1 \cdot \tan\phi_2 = 1$
- C) $\tan\phi_1 \cdot \tan\phi_2 = -1$ D) $\tan\phi_1 + \tan\phi_2 = 1$
12. The value of c got by applying Lagrange's mean value theorem to the function $f(x) = e^x$ in $[0, 1]$ is _____
- A) $\log(e-1)$ B) e
- C) 1 D) $e-1$
13. If $u = x^3 + y^3$, then $\frac{\partial^2 u}{\partial x \partial y}$ is _____
- A) 3 B) 0
- C) $3x + 3y$ D) -3
14. If $r = f_{xx}, s = f_{xy}, t = f_{yy}$, then $f(x, y)$ has minimum at (a, b) if,
- A) $rt - s^2 > 0$ & $r < 0$ at (a, b) B) $rt - s^2 < 0$ & $r < 0$ at (a, b)
- C) $rt - s^2 > 0$ & $r > 0$ at (a, b) D) $rt - s^2 < 0$ & $r > 0$ at (a, b)
15. Given $u = x + y, v = xy$ then $\frac{\partial(u,v)}{\partial(x,y)} =$ _____
- A) $x + y$ B) 0
- C) $x - y$ D) 1
16. If $u = y^x$ then $\frac{\partial u}{\partial x} =$ _____
- A) $y^x \log y$ B) 1
- C) xy^{x-1} D) $y^x \log x$
17. The value of $\int_0^{\pi/2} \int_0^2 r dr d\theta =$ _____
- A) $\frac{\pi}{4}$ B) π
- C) $\frac{\pi}{2}$ D) $\frac{3\pi}{4}$
18. Which of the following formula can be used to find area of a plane region in the cartesian form?
- A) $A = \iint x dx dy$ B) $A = \iint xy dx dy$
- C) $A = \iint f(x, y) dx dy$ D) $A = \iint dx dy$
19. Which of the following is true?
- A) $\Gamma(n+1) = n\Gamma(n)$ B) $\Gamma(n-1) = \Gamma(n)$
- C) $\Gamma(n+1) = n + 2\Gamma(n)$ D) $\Gamma(n+2) = n\Gamma(n)$
20. The value of $\beta(3, 2)$ is:
- A) $1/12$ B) $1/2$
- C) $1/10$ D) $1/24$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Transform following matrix to echelon form and hence find rank of the matrix $\begin{bmatrix} 1 & 2 & 8 \\ 4 & 5 & 8 \\ 3 & 2 & 1 \end{bmatrix}$.	4	L*2	1	1
b)	Solve the system of equation by LU decomposition method $x + y - z = 2$, $2x + 3y + 5z = -3$ and $3x + 2y - 3z = 6$.	6	L3	1	1
c)	Apply Gauss Seidel method and obtain solution of system of equations $27x + 6y - z = 85$, $6x + 15y + 2z = 72$ and $x + y + 54z = 110$. Carryout 3 iterations.	6	L3	1	2
2. a)	By reducing to echelon form show that rank the following matrix is 2. $\begin{bmatrix} 1 & 2 & -1 & 3 \\ 3 & 4 & 0 & -1 \\ -1 & 0 & -2 & 7 \end{bmatrix}$	4	L2	1	1
b)	Using Thomas algorithm solve the system of equations $\begin{bmatrix} 4 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}$.	6	L3	1	1
c)	Test for consistency and solve the system of equations $x + y + z = 9$, $2x - 3y + 4z = 13$ and $3x + 4y + 5z = 40$. by Gauss elimination method.	6	L3	1	2
Unit – II					
3. a)	Build the characteristic equation and hence find the Eigenvalues of the matrix $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$.	4	L2	2	1
b)	Apply power method and find largest Eigenvalue and Eigenvector of the matrix $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ by taking the initial approximation to the eigen vector as $[1 \ 0 \ 0]^T$.	6	L3	2	1
c)	Diagonalize the following matrix $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.	6	L3	2	2
4. a)	Find the Eigenvalues and Eigen vectors of the matrix $\begin{bmatrix} 4 & -5 \\ 1 & -2 \end{bmatrix}$.	4	L2	2	1
b)	Apply Given's method to transform symmetrix to tridiagonal form. $\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$	4	L3	2	1

- c) Apply the Jacobi method to compute the Eigenvalues of the

$$\text{symmetric matrix } \begin{bmatrix} 2 & \sqrt{2} & 4 \\ \sqrt{2} & 6 & \sqrt{2} \\ 4 & \sqrt{2} & 2 \end{bmatrix}$$

8 L3 2 2

Unit – III

5. a) Obtain the Maclaurin's series expansion for the following

$f(x) = \log(1+x)$ upto 3 nonzero terms.

4 L3 3 1

- b) Verify Lagrange's theorem for the following function. $f(x) = e^x$ in $[0,1]$.

6 L2 3 1

- c) Determine the angle between the curves $r = a(1 + \cos\theta)$ and $r = b(1 - \cos\theta)$.

6 L3 3 2

6. a) Expand $\sin x$ in Taylor's series expansion with period $(x - \pi/2)$ up to terms containing 4th degree term.

4 L3 3 1

- b) Show that with usual notation $\tan\phi = r \frac{d\theta}{dr}$

6 L2 3 1

- c) Determine radius of curvature of the cardioid $r = a(1 + \cos\theta)$,

6 L3 3 2

Unit – IV

7. a) If $z = f(x+ct) + \phi(x-ct)$, then prove that $\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$.

4 L2 4 1

- b) If $u = x + \frac{y^2}{x}$, $v = \frac{y^2}{x}$ then evaluate the Jacobian $\frac{\partial(y,v)}{\partial(x,y)}$

6 L3 4 1

- c) Obtain the extreme values of $u = x^2 + y^2 + 6x - 1$

6 L3 4 2

- 8 a) If $xy^3 - yx^3 = 6$ is the equation of the curve then find the slope and the equation of the tangent line at (1, 2).

4 L2 4 1

- b) If $u = f(2x-3y, 3y-4z, 4z-2x)$, show that

$$\frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z} = 0.$$

6 L3 4 1

- c) Find the maximum value of $x^2 + y^2 + z^2$ when $x + y + z = 3a$ by Lagrange's multiplier method.

6 L3 4 2

Unit – V

9. a) Evaluate $\int_1^2 \int_0^3 xy^2 dx dy$

4 L3 5 1

- b) Prove that $\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$

6 L2 5 1

- c) Determine the area bounded by the cardioid $r = a(1 + \cos\theta)$.

6 L3 5 2

- 10 a) Evaluate $\int_0^{\pi/2} \sqrt{\tan\theta} d\theta$ using Beta & Gamma function.

4 L3 5 1

- b) Show that $\int_0^1 \int_0^2 \int_0^3 (4xyz) dx dy dz = 18$.

6 L2 5 1

- c) determine the volume common to the cylinders $x^2 + y^2 = a^2$ and $x^2 + z^2 = a^2$

6 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
